✅ 1. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions and equations, and geometric representations

- Highest relevance.
- The core of this task lies in manipulating an algebraic formula:
  - Students must rearrange $D = \frac{dv}{60n}$ to obtain $v = \frac{60nD}{d}$;
  - Alternatively, they may reason numerically without explicit rearrangement (e.g., "first find total drops, then convert to volume").
- It also involves numerical operations: integer multiplication and division (60 × 3 × 50 ÷ 25).
- This demonstrates a blend of structural understanding of algebra and computational fluency, making Competency #7 the most directly engaged.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving requires a clear sequence of steps. For example:Method A (Algebraic rearrangement):

  1. Rearrange the formula to solve for v ;
  2. Substitute $D = 50$, $n = 3$, $d = 25$;
  3. Compute $v = \frac{60 \times 3 \times 50}{25} = 360$ mL.

  Method B (Contextual reasoning):

  1. Calculate total drops: 50 drops/min × 60 min/hour × 3 hours = 9000 drops;
  2. Convert to volume: 9000 drops ÷ 25 drops/mL = 360 mL.
- Both approaches involve a logically coherent multi-step process, fully aligning with Competency #12.

✅ 3. Competency #1: Perform a simple calculation

- Highly relevant as a foundational skill.
- Although the arithmetic is straightforward (multiplication and division), accuracy is critical—especially in a medical context.
- Key computation:

$$\frac{60 \times 3 \times 50}{25} = \frac{9000}{25} = 360$$

- Errors in unit conversion (minutes vs. hours) or division could lead to serious consequences. Thus, reliable execution of basic operations is essential.

✅ 4. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Implicitly expected.
- While the prompt doesn't explicitly say "show your work," in educational or clinical settings, presenting key steps (e.g., formula rearrangement or total drop count) helps verify correctness.
- A strong response should include:
  - The chosen method or formula;
  - Intermediate results (e.g., "total drops = 9000");
  - Final answer with units ("$v = 360$ mL").
- This transparency reflects Competency #8.

✅ 5. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

- Moderately relevant.
- The medical context provides important cues:
  - Calculating "volume from drip rate" is routine for nurses; students can use real-world logic to check plausibility (e.g., 360 mL over 3 hours is a common infusion volume);
  - Unit consistency (minutes vs. hours) must be handled carefully—the factor of 60 in the formula exists precisely to reconcile time units, and the context reminds students to verify unit alignment.
- Thus, professional contextual knowledge supports both modeling and validation.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 本题核心在于操作代数公式：
  - 学生需将公式 $D = \frac{dv}{60n}$ 变形为 $v = \frac{60nD}{d}$；
  - 或在不显式变形的情况下，通过逆向算术推理（如“先算总滴数，再除以滴系数”）求解。
- 同时涉及数值操作：处理整数乘除（60 × 3 × 50 ÷ 25）。
- 这体现了对代数结构的理解与数值计算的结合，是能力 #7 的典型应用。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需清晰的多阶段流程，例如：
  方法一（代数变形）：

  1. 从 $D = \frac{dv}{60n}$ 解出 v；
  2. 代入 $D = 50, n = 3, d = 25$；
  3. 计算 $v = \frac{60 \times 3 \times 50}{25} = 360$ mL。

  方法二（情境推理）：

  1. 计算总滴数：50 滴/分 × 60 分/小时 × 3 小时 = 9000 滴；
  2. 转换为体积：9000 滴 ÷ 25 滴/mL = 360 mL。
- 两种路径都包含逻辑连贯的多个步骤，完全符合能力 #12。

✅ 3. 能力 #1：执行简单计算

- 高度相关（作为支撑）。
- 虽然计算本身不复杂（乘法与除法），但准确性至关重要（医疗场景）。
- 关键计算：

$$\frac{60 \times 3 \times 50}{25} = \frac{9000}{25} = 360$$

- 若学生在单位换算（分钟↔小时）或除法上出错，将导致严重后果。因此，可靠地执行基本运算是本题成功的基础。

✅ 4. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 隐含要求。
- 虽然题目未明确说“展示过程”，但在教育或临床评估中，展示关键步骤（如公式变形或总滴数计算）有助于验证思路正确性。
- 优秀回答应包括：
  - 公式或推理路径；
  - 中间结果（如“总滴数 = 9000”）；
  - 带单位的最终答案（“v = 360 mL”）。
- 这种透明表达体现了能力 #8。

✅ 5. 能力 #13：利用对情境的理解来指导或加速数学解题过程

- 中等相关。
- 医疗情境提供重要线索：
  - “滴速 → 总量”是护士日常工作，学生可借助现实逻辑验证答案合理性（如 360 mL 在 3 小时内输完是常见剂量）；
  - 单位一致性（分钟 vs. 小时）需特别注意——公式中的 60 正是为了统一时间单位，情境提醒学生检查单位匹配。
- 因此，专业背景知识支持了建模与验算。